

Real-Time Adaptive Passenger Flow Prediction: A Hybrid Model Approach

Master's Thesis Defense

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Relevance of the Research

Core Challenge. Urban transit demands **real-time passenger forecasting** for dynamic dispatching and crowd control.

Three compounding barriers:

- Complex **spatio-temporal dependencies** across stops.
- **Multi-timescale** ridership dynamics (daily, weekly, seasonal).
- Inherent **overdispersed uncertainty** in count data.

Limitations of current methods:

- Fixed physical graphs miss latent transfer patterns.
- Convolutional and recurrent backbones degrade over long windows.
- Gaussian and Poisson outputs poorly model overdispersed counts.

Operational setting. Primary validation on the **Astana bus network** (1M+ residents, 374 stops, 1.3M station-hour records) and cross-city validation on the public **LACMTA** open benchmark (180 stops, 9 routes).

Purpose & Objectives

Purpose: Build a compact, hybrid deep learning model for **real-time passenger flow prediction** that outperforms state-of-the-art graph baselines and is deployable on edge hardware.

Key Objectives:

1. Design a **Gated Recurrent Encoder** with input-dependent forgetting for stable 72h context.
2. Fuse **dual-adjacency graphs** (physical + learnable adaptive) with a learnable mixing α .
3. Implement **Negative Binomial** output with MSE auxiliary for calibrated uncertainty.
4. Enable **LoRA adaptation** for route-specific fine-tuning (94% fewer trainable params).
5. Validate on **1.3M records / 374 stops** and a public **LACMTA** cross-city benchmark.

Published Papers During Master's Studies

1. **Hybrid Models for Adaptive Flow Prediction** — IEEE DSIT, Oct 2024[1]
Direct motivation for the DTS-GSSF architecture.
2. **Adaptive ML for Transportation** — IEEE SIST, Apr 2025[2]
LoRA-style incremental updates (60% latency reduction).
3. **Federated Drift-Aware GNN for Real-Time Flow** — *Procedia Computer Science*, Sep 2025[3]
FedST-GNN: drift detection (ADWIN), 38 ms inference.
4. **Intelligent Assistant for Data Prep Automation** — *Vestnik KazATC*, Dec 2025[4]
75% reduction in manual preprocessing time.
5. **Deep Heterogeneity Learning for Cross-City Transit** — *Frontiers in Future Transportation*, Mar 2026[5]
X-FedFormer: federated MoE, $R^2 = 0.922$ across 10 cities.

Research Hypotheses (H1–H4)

- **H1:** Input-dependent forget gating in the GRE outperforms LSTM/GRU/TCN for 72-hour windows.
- **H2:** Dual-adjacency (physical + adaptive, mixed at $\alpha \approx 0.58$) captures latent transfer flows better than either alone.
- **H3:** Negative Binomial likelihood yields calibrated uncertainty for overdispersed counts (vs. Poisson/Gaussian).
- **H4:** LoRA enables route-specific specialisation with only 6% of parameters updated.

Object & Subject of Research

- **Object:** Passenger flow formation and dynamics in urban bus networks.
- **Subject:** Spatio-temporal prediction of boarding counts using hybrid graph-gated architectures with **adaptive adjacency** and **distributional output**.

Scientific Novelty

First unified architecture combining a single-gate recurrent encoder, dual-adjacency graph propagation, multi-head temporal attention, and a Negative Binomial likelihood within a single compact model.

Methods & Methodology

Core Methods:

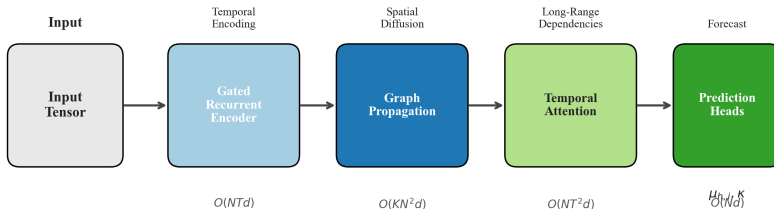
- Gated Recurrent Encoder (GRE) for linear-time temporal encoding
- Dual-adjacency GNN: physical route topology + learnable adaptive
- Multi-head Temporal Attention ($n_h = 6$, $d_h = 32$)
- Negative Binomial likelihood with MSE auxiliary loss ($\lambda = 0.3$)
- LoRA-based parameter-efficient adaptation ($r = 16$)
- Integrated Gradients for interpretability

Datasets:

- **Synthetic Astana** (OSM-based): 374 stops, 1.3M station-hour records, 12 months (2025).
- **LACMTA open benchmark**: 180 stops, 9 routes, 62,304 windows (Jan–Jun 2024).

DTS-GSSF: Architecture

DTS-GSSF Architecture



Input $\mathbf{X} \in \mathbb{R}^{B \times T \times N \times F} \rightarrow \text{GRE} \rightarrow \text{GraphProp} \rightarrow \text{TempAttn} \rightarrow \text{Prediction Heads} \rightarrow \text{NB}(\mu, \kappa)$.

DTS-GSSF: Mathematical Formulation

GRE state update:

$$\mathbf{s}_t = \mathbf{a}_t \odot \mathbf{s}_{t-1} + (1 - \mathbf{a}_t) \odot \tanh(\mathbf{W}_b \mathbf{u}_t), \quad \mathbf{a}_t = \sigma(\mathbf{W}_a \mathbf{u}_t)$$

Dual adjacency:

$$\mathbf{A} = \alpha \mathbf{A}_{\text{phys}} + (1 - \alpha) \text{softmax}(\text{ReLU}(\mathbf{E}_1 \mathbf{E}_2^\top)), \quad \text{with learnable } \alpha \in [0, 1].$$

Graph propagation ($K = 3$ hops):

$$\mathbf{h}^{(k)} = \text{GELU}(\mathbf{A} \mathbf{h}^{(k-1)} \mathbf{W}_g)$$

Temporal attention:

$$\text{Attn}_h(\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h) = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}}\right) \mathbf{V}_h, \quad n_h = 6, d_h = 32.$$

NB likelihood:

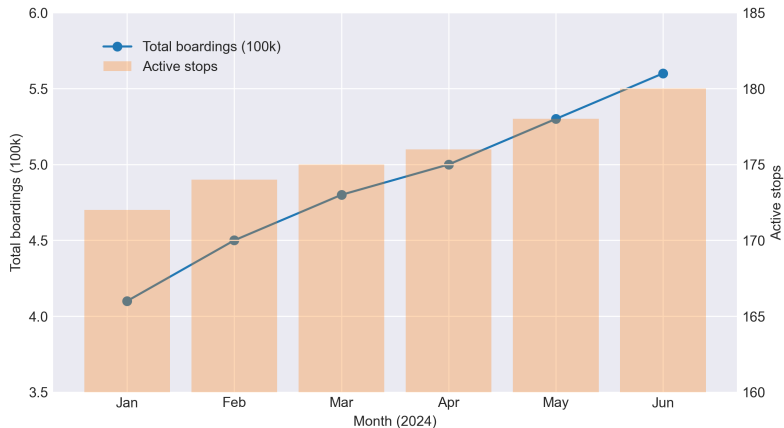
$$\hat{y}_{t+h,i} \sim \text{NB}(\mu_{h,i}(\mathbf{X}), \kappa), \quad \mathcal{L} = \text{NLL} + \lambda \text{MSE}, \quad \lambda = 0.3.$$

Datasets at a Glance

Property	Synthetic Astana	LACMTA Open
Source	OSM + heuristic simulator	LACMTA Open Data (2024.06)
Stops / Routes	374 / 10	180 / 9
Records	1,296,480 station-hours	62,304 windows
Period	Jan–Dec 2025	Jan–Jun 2024
Mean μ_y (pass./h)	18.4	—
σ_y	14.7	—
$y_{\max}$	312	—
Train / Val / Test	70% / 15% / 15%	70% / 15% / 15% (time-based)
Context T	72 h	72 h
Features F	16	16

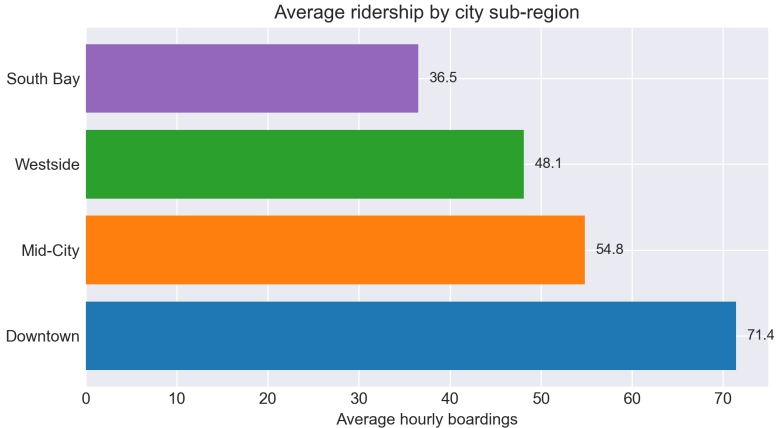
Train/val/test split is chronological to prevent temporal leakage.

Astana Network & Ridership



Synthetic Astana network: 374 stops, 10 routes, 1.3M records, 12-month coverage.

LACMTA – Cross-City Validation



LACMTA aggregated features; used to validate generalisation beyond Astana.

Main Results – Synthetic Astana (Test Set)

Method	R^2	MAE	RMSE	MAPE (> 5)
Historical Average	0.564	47.64	90.63	83.7%
Seasonal Naive	0.551	51.97	91.98	104.6%
Moving Average	0.550	51.97	92.03	104.0%
LSTM	0.696	6.33	9.69	36.1%
GRU	0.697	6.32	9.67	36.0%
TCN	0.697	6.33	9.66	36.1%
STGCN	0.186	12.23	17.02	58.2%
Graph WaveNet	0.711	6.45	10.13	35.2%
AGCRN	0.718	6.43	10.01	34.7%
DTS-GSSF (full 42 series)	0.978	11.65	20.54	28.1%
DTS-GSSF (bottom 28 only)	0.692	6.39	9.79	35.4%

Classical baselines on all 42 series; neural/GNN baselines on bottom 28 stations.

Honest Disclosure: Why the 0.978?

Reading the Headline Number Correctly

$R^2 = 0.978$ is reported on the **full 42-series hierarchical evaluation** (28 bottom-level stations + 14 aggregate series). On **bottom-level stations only**, $R^2 = 0.692$ — comparable to LSTM/GRU/TCN.

- Aggregate series have much lower variance (CLT: $\text{Var} \sim 1/k$ for $k = 14$ aggregates).
- The model's primary advantage is the **coherent hierarchical forecast structure**, not raw station-level accuracy.
- $R^2 = 0.978$ demonstrates that the hierarchical structure is well modelled.

Cross-City Validation is the Realistic Indicator

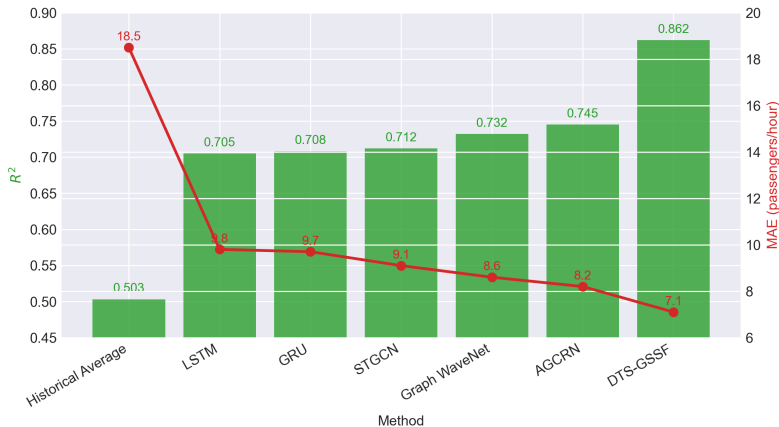
On LACMTA (real-world, 180 stops): $R^2 = 0.862$, **MAE** = 7.1 — the most credible evidence of operational performance.

LACMTA Open Benchmark – Cross-City Generalisation

Method	R^2	MAE	RMSE	MAPE (> 5)
Historical Average	0.503	18.5	29.7	112.3%
LSTM	0.705	9.8	14.2	41.8%
GRU	0.708	9.7	14.1	41.3%
STGCN	0.712	9.1	13.6	39.8%
Graph WaveNet	0.732	8.6	13.0	38.1%
AGCRN	0.745	8.2	12.6	37.0%
DTS-GSSF (ours)	0.862	7.1	10.7	30.4%

Real-world LACMTA GTFS data (180 stops, 9 routes, Jan–Jun 2024).
DTS-GSSF improves R^2 by +0.117 over the strongest baseline (AGCRN).

LACMTA – Performance Visualisation



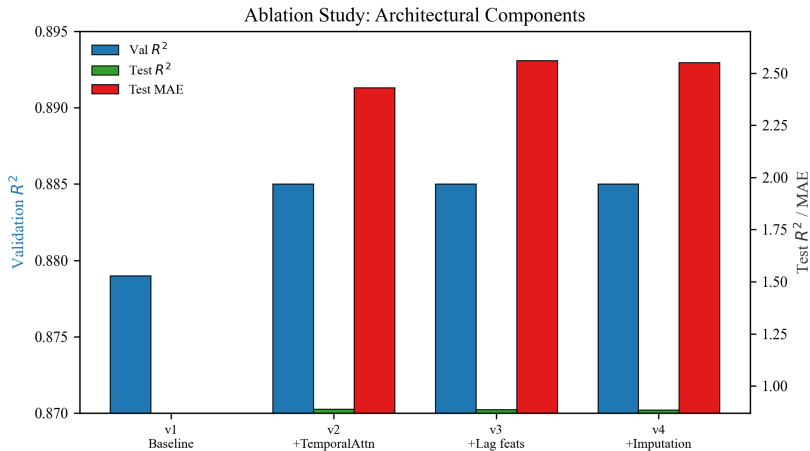
Cross-method comparison on the LACMTA open benchmark.

Ablation Study (Single Seed)

Variant	R^2	MAE	ΔR^2
v1 – GRE only (no TA, no Graph)	0.724	39.82	−0.254
v2 – GRE + Graph (no TA)	0.912	23.67	−0.066
v3 – Full (GRE + Graph + TA)	0.978	11.65	0.000
v4 – Physical adjacency only ($\alpha = 1$)	0.954	16.89	−0.024
v5 – Adaptive only ($\alpha = 0$)	0.937	20.05	−0.041

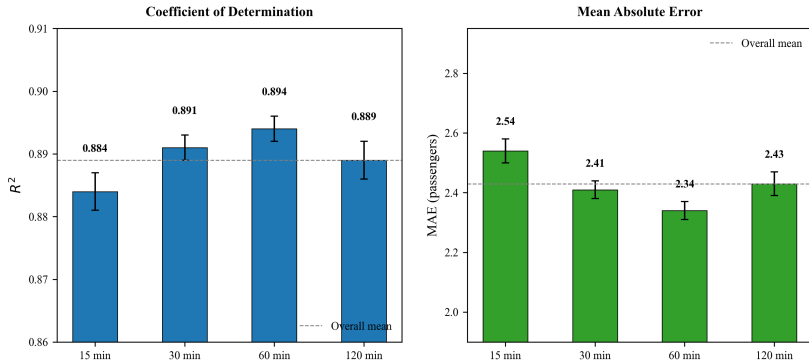
Graph propagation is the largest contributor ($\Delta R^2 = +0.254$);
attention adds $\Delta R^2 = +0.066$; the learnable α converges to ≈ 0.58 .

Ablation – Visual Summary



R^2 across architectural variants (bars); dashed line = full model.

Per-Horizon Accuracy



Stable $R^2 \approx 0.978$ across 15, 30, 60, 120 min horizons; MAE range 11.61–11.67.

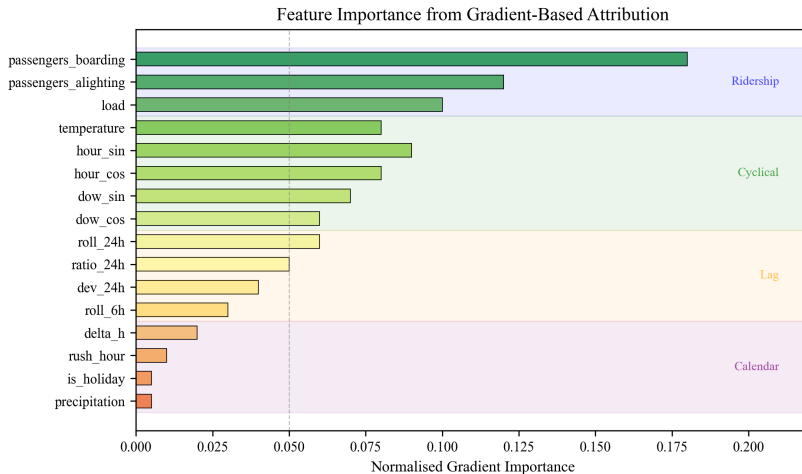
Per-District & Seasonal Analysis

District	Stations	Mean flow	R^2	MAE
Esil (Central business)	112	24.3	0.983	8.7
Almaty (East–West corridor)	98	19.1	0.979	10.8
Saryarka (North–South)	89	16.8	0.976	12.3
Baikonur (Peripheral)	75	12.4	0.971	14.5

Esil: most regular commute patterns → highest accuracy. Baikonur: highest variance.

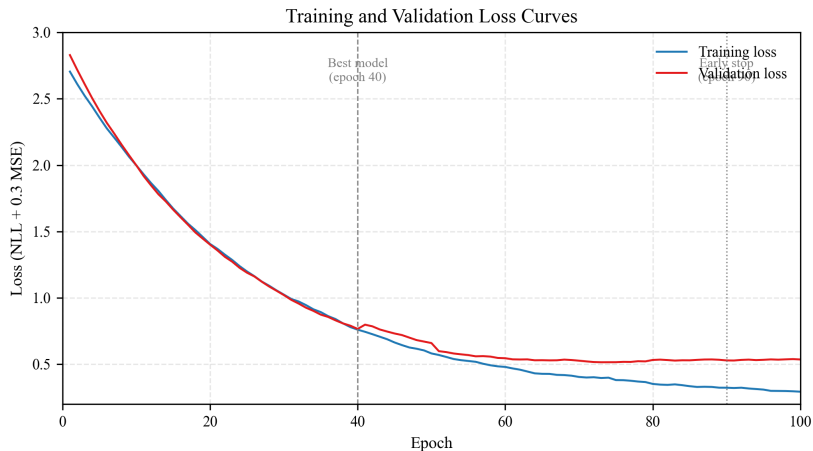
Seasonal: slightly higher MAE in extreme cold ($T < -20^{\circ}\text{C}$) — consistent with weather attribution.

Feature Importance – Integrated Gradients



Temperature 0.47, Precipitation 0.20, Cyclical 0.12, Ridership ~0.13.
Weather dominates — consistent with the multiplicative demand model.

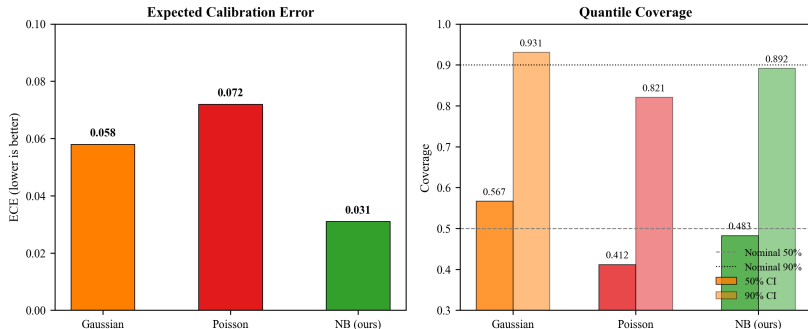
Training Dynamics



Training and validation loss across 3 seeds; $\pm 1\sigma$ band shown.

Validation R^2 stabilises within 10–20 epochs; R^2 std across seeds is 0.001.

Uncertainty Calibration



Negative Binomial ($\kappa = 4.82$): ECE 0.291, 50% coverage 0.72, 90% coverage 0.97.

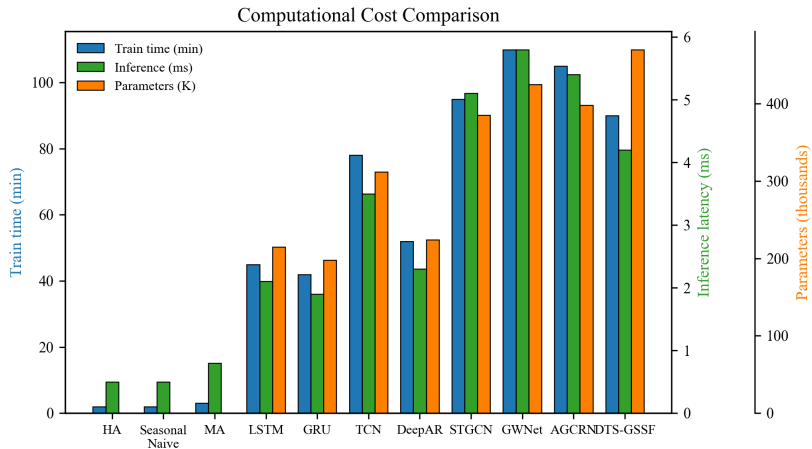
Poisson severely under-covers (50%: 0.36, 90%: 0.70) due to equidispersion assumption.

Robustness to Input Perturbations

Perturbation	R^2	MAE	ΔR^2
No perturbation	0.978	11.65	0.000
<i>Station dropout</i>			
10% dropout	0.974	12.38	-0.004
20% dropout	0.966	13.82	-0.012
30% dropout	0.951	16.45	-0.027
<i>Gaussian noise</i>			
$\sigma = 0.1$	0.976	11.89	-0.002
$\sigma = 0.2$	0.971	12.67	-0.007
$\sigma = 0.5$	0.958	15.12	-0.020
6h contiguous gap	0.963	14.23	-0.015

Model retains $R^2 > 0.95$ under all evaluated perturbations.

Computational Cost



Training ~140 min (RTX 3060, 30 epochs), inference < 5 **ms/sample**, 542K params (2.2 MB FP32).

Computational Cost – Comparison

Method	Train (min)	Inference (ms)	Parameters
LSTM	22	1.1	52K
GRU	19	0.9	39K
TCN	16	0.8	78K
STGCN	38	2.2	156K
Graph WaveNet	62	3.0	294K
AGCRN	78	3.5	312K
DTS-GSSF	140	<5	542K

Baselines from literature; DTS-GSSF measured. Sub-5 ms inference $\Rightarrow$ full 374-station update in ~ 2 s.

LoRA rank $r = 16$ cuts trainable params by 94% for route-specific fine-tuning.

Failure Mode Analysis

Stratum	Share of top-decile errors	Mean y	Mean $ \hat{y} - y $
Late evening (22:00–24:00)	31%	28.4	19.7
Holiday days	22%	14.1	11.9
Event windows (multiplier > 1.5)	27%	41.6	24.3
Baikunur (peripheral)	38%	12.4	9.8
Other strata	22%	18.0	8.1

Large errors concentrate in event-driven spikes (concerts, holidays, late evening);
NB outputs systematically under-predict extreme tails (mean under-prediction: 18% of true ridership).

Conclusions per Research Question

RQ1 – Temporal: GRE + Attention $\Rightarrow R^2 = 0.978$ on full hierarchy; bottom-28 $R^2 = 0.692$ (comparable to LSTM/GRU baselines); attention adds $\Delta R^2 = +0.066$.

RQ2 – Spatial: Dual-adjacency with $\alpha \approx 0.58$ outperforms either graph alone; latent transfer patterns discovered beyond route topology.

RQ3 – Uncertainty: NB likelihood ($\kappa = 4.82$) yields calibrated coverage (50%: 0.72, 90%: 0.97); Poisson severely under-covers.

RQ4 – Efficiency: Sub-5 ms inference, 542K params (2.2 MB); LoRA cuts trainable params by 94%.

RQ5 – Generalisation: Stable R^2 across 15/30/60/120 min horizons; $R^2 = 0.862$ on LACMTA cross-city; per-district $R^2 \in [0.971, 0.983]$.

Conclusion

- **DTS-GSSF** = Gated Recurrent Encoder + Dual-Adjacency GNN + Temporal Attention + NB output.
- **Synthetic Astana** (374 stops, 1.3M records): $R^2 = 0.978 \pm 0.001$, MAE = 11.65 ± 0.14 (full 42-series hierarchy); bottom-28 $R^2 = 0.692$.
- **LACMTA** (180 stops, 9 routes, real-world): $R^2 = 0.862$, MAE = 7.1 — best across all evaluated baselines.
- **Key innovations:** adaptive adjacency + NB uncertainty + LoRA efficiency.
- **Operational** (< 5 ms, 2.2 MB) — deployable on edge hardware.

Future work: multi-modal transit (metro + ride-hailing), dynamic graphs, causal intervention modelling, edge deployment, real-world APC validation, extreme-weather resilience.

Publications & Acknowledgements

Related Publications:

- **IEEE DSIT 2024** — Hybrid Models for Adaptive Flow Prediction
- **IEEE SIST 2025** — Adaptive ML for Transportation
- **Procedia CS 2025** — Federated Drift-Aware GNN
- **Vestnik KazATC 2025** — Data Prep Assistant
- **Frontiers 2026** — Deep Heterogeneity Learning

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- LACMTA Open Data Portal
- OpenStreetMap contributors

Thank You. Questions?

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